Filipe Nunes Vicente

Research Scientist in Cell Biology and Imaging

Address: Felix-Wankel Strasse 19, 69126 Heidelberg, Germany
Email: filipenunesvicente1@gmail.com **| Contact:** +33695702853 **| ORCID:** 0000-0001-5107-0461
Linkedin: filipe-nunes-vicente | **GitHub:** fnunesv | **Website:** https://fnunesv.github.io/

SUMMARY

I am an interdisciplinary cell biologist and biophysicist with 10 years of experience in cellular and molecular biology, advanced fluorescence microscopy, and the development of quantitative assays. I am passionate about designing synthetic molecular tools to regulate cellular functions and developing high-throughput imaging methods.

EXPERIENCE

Postdoctoral researcher **01/2022-Present**

Diz-Muñoz Lab, EMBL Heidelberg, Germany

- Developed three main GLP-compliant workflows for engineering multiclonal and single clonal cell lines:
    - 3T3 fibroblasts expressing inducible or constitutive custom-designed optogenetic actin crosslinkers coupled to signalling biosensors or tagged cytoskeletal proteins.
    - MEF fibroblasts with CRISPR knockouts for the membrane skeletal protein spectrin.
    - MDCK epithelial cells and W256 carcinosarcoma cells expressing inducible optogenetic actin crosslinkers.
- Designed six different optogenetic CRY2 or iLID-based actin crosslinkers with scalable molecular lengths, enabling nanometric control of actin spacing from 10 to 50nm.
- Performed molecular and phenotypic characterisation of engineered cell lines through PCR, Western Blot, flow cytometry, bulk RNA-seq, and automated live cell imaging.
- Established 3D cell culture and imaging of MDCK epithelial multicellular cysts up to 10 days of maturation.
- Designed a magnetic bead assay coupled to optogenetics to measure material properties of live cells with nanometric precision and time resolution of ~1s.
- Developed and implemented two high-throughput automated workflows for microscopy:
    - Feedback Airyscan microscopy 2D and 3D timelapses coupled to single-cell optogenetic activation of up to 20 cells per sample.
    - Automated Nikon SoRA confocal spinning disk microscopy for 3D imaging of cytoskeletal proteins in hundreds of fixed cells per sample.
- Designed three AI-driven image analysis pipelines to segment, quantify and classify microscopy data for more than 20 different morphodynamic parameters.
- Supervised three MSc students and collaborated with international teams of physicists, biologists, and chemists.
- Presented work in several interdisciplinary international conferences and partner labs.

Predoctoral researcher and scientist **09/2016-09/2021**

Giannone Lab, IINS Bordeaux, France

- Developed two custom-microfabricated 3D-printed devices to combine for the first time single and cyclic cell stretching with super-resolution microscopy and single particle tracking.
- Implemented long-term sterile culture of primary hippocampal neurons and chicken spinal cord explants in custom-microfabricated stretching devices.
- Applied stretching devices to study and model biological processes in various cell types expressing fluorescent-tagged target proteins:
    - Cell-matrix adhesion and cytoskeletal organisation in fibroblasts (MEF).
    - Membrane structures in endothelial cells (MLEC).
    - Axonal injury in primary neurons and chicken spinal cord explants.
- Optimised click-chemistry probes for super-resolution microscopy of dendritic spines in primary neurons.
- Published two first-author articles in high-impact-factor journals and collaborated with four other publications.
- Supervised one MSc and one PhD student and collaborated with interdisciplinary research teams in physics, optics, and chemistry.
- Presented work in several interdisciplinary international conferences and partner labs.

Grants and awards

- Joachim Herz Add-On Fellowship for Interdisciplinary Life Sciences: 12500 € (01/2023-01/2026)
- EMBO Postdoctoral Fellowship: full postdoctoral salary coverage (01/2022-01/2024)
- French MCESR PhD grant: PhD salary coverage (09/2016-09/2019)

Education

University of Bordeaux, France **09/2016-11/2020**
PhD in Cell Biology and Neuroscience (Thesis supervisor: Dr. Grégory Giannone)

University of Coimbra (Portugal) /Neurocentre Magendie(France) **09/2014-07/2016**
MSc. in Cellular and Molecular Biology (MsC supervisors: Giovanni Marsicano and Román Serrat)

University of Coimbra (Portugal) **09/2011-07/2014**
B.Sc. in Biochemistry

Skills:

- **Cell and tissue culture:** adherent cell lines (fibroblasts, epithelial, endothelial); suspension cell lines (carcinosarcoma), primary hippocampal neurons; tissues (chicken spinal cord explants).
- **Cellular and molecular biology:** stable cell lines and single cell clones; transfection (electroporation, lentiviral infection, lipofection); flow cytometry; immunocytochemistry; CRISPR knockouts; plasmid design (Gibson Assembly, site-directed mutagenesis, PCR-based cloning); Western Blot.
- **Optical microscopy:** super-resolution widefield and confocal (PALM, STORM, DNA-PAINT, STED); single particle tracking; confocal spinning-disk and AiryScan; phase contrast; adaptive feedback microscopy.
- **Image analysis:** machine learning image analysis and segmentation (Fiji, CellPose, Napari, Ilastik); cell and single molecule tracking (Imaris, Metamorph, TrackMate); multidimensional image processing and visualization.
- **Material science and bioengineering:** Atomic force microscopy; magnetic-bead rheology; microfabrication; 3D-printing and milling; 3D cell culture and elastic substrates (collagen gels, PDMS assemblies); micropatterning.
- **Programming:** R (advanced); Python (intermediate); ImageJ macro language (proficient); Claude Code; Git.
- **Software:** image edition (Illustrator, Photoshop); computer-assisted design (FreeCAD, Solidworks), imaging acquisition software (NIS Elements, LAS X, ZEN, Metamorph); flow cytometry (FlowJo); protein modelling (AlphaFold, ChimeraX); ELN (eLabJournal); project management and tracking (Notion).

Selected Publications

Massou, S.#, **Nunes Vicente, F.#,** Wetzel, F.#, Mehidi, A., Strehle, D., Leduc, C., Voituriez, R., Rossier, O., Nassoy, P.,Giannone, G., 2020. Cell stretching is amplified by active actin remodeling to deform and recruit proteins in mechanosensitive structures. Nature Cell Biology 22, 1011–1023. https://doi.org/10.1038/s41556-020-0548-2
#Shared first authorship.

Nunes Vicente, F.*, Lelek, M., Tinevez, J.-Y., Tran, Q.D., Pehau-Arnaudet, G., Zimmer, C., Etienne-Manneville, S., Giannone, G., Leduc, C., 2022. Molecular organization and mechanics of single vimentin filaments revealed by super-resolution imaging. Science Advances 8, eabm2696. https://doi.org/10.1126/sciadv.abm2696
*First author.

All publications can be viewed at: https://fnunesv.github.io/publications/

Teaching

Practical Course Instructor, EMBL Heidelberg, Germany **09/2024; 09/2022**
EMBO Current Methods in Cell Biology: atomic force microscopy, micropatterning

Practical Course Instructor, University of Bordeaux, France **09/2018**
Cajal Neuroscience Summer School: super-resolution microscopy in neurons

Supervision experience

Yogesh Pratap, master student, University of Bayreuth **09/2025-12/2025**
Célia Galleri-Paris, master student, ENS Lyon **01/2025-07/2025**
Sara Fernández Hermira, master student, University of Amsterdam **09/2022-07/2023**
Xuesi Zhou, PhD student, University of Bordeaux **01/2021-07/2021**
Marion Boyer, master student, Université de Bordeaux **01/2021-07/2021**

Professional training

Lab Leadership for Scientists, EMBO, Heidelberg, Germany **11/2024**
Project Management for Scientists, EMBO (online) **10/2024**
EMBO Practical Course in Computational Optical Biology, IGC Lisbon, Portugal **10/2022**
Frontiers in Neurophotonics Summer School, CERVO, Quebec City, Canada **07/2018**

Community involvement and volunteering

Contributor, PreLights, The Company of Biologists **05/2026-Present**
The biology preprint highlighting service hosted by The Company of Biologists
Translator, Scientific Outreach Team, EMBL Heidelberg, Germany **04/2026-Present**
High school resources in epigenetics
Member, CoMaking Space, Heidelberg, Germany **01/2022-Present**
Open-access 3D-printing and milling, woodworking, and electronics
Member, Graines de Solidarité, Bordeaux, France **09/2020-09/2021**
Non-profit organisation dedicated to providing food support to people and families in need
French tutor, Consulat du Portugal, Bordeaux, France **04/2019-09/2021**
Reporter, Rádio Universidade de Coimbra, Portugal **09/2014-09/2015**
Hosting live broadcasts and interviews, news redaction, and creation of a weekly podcast about fake news

Languages

Portuguese (native); English (proficient); French (advanced); Spanish (advanced); German (elementary)

Interests

Sports (Muay Thai, Boxing, Football); Gardening; Cooking; Crafting (3D-printing, woodworking); Neuroscience; Reading (economics, spirituality, AI, historical & science fiction)